

(Please write your Enrollment Number)

Enrollment No. _____

REMID-TERM EXAMINATION
(Course Name: B.Tech/) (Semester: I)
(November 2024) OFF LINE mode

Subject Code: BAI 102	Subject: IT Workshop
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

		(2.5*4=10)	CO Mapping
Q1			CO1
a)	What is the purpose of assignment operators in R? Discuss the difference between <-, =, and ->		CO1
b)	What are the special values in R, such as NA, NaN, Inf, and NULL? How do they differ?		CO2
c)	Explain the difference between numeric and character vectors. How are they created in R? What is vector recycling in R? How does it work in arithmetic operations?		CO2
d)	How do you bin continuous data into categories in R? Why is binning useful in data analysis?		
UNIT I			
Q2	Attempt any two parts	(5*2=10)	CO Mapping
a)	Create a vector with missing value (NA) and calculate its mean. Handle missing values appropriately. What is the purpose of the is.na() function? How do you handle missing data in R?		CO1
b)	What are R packages, and why are they important? How do you use a package in R? Please explain the different steps involved.		CO1
c)	What is the difference between base R and contributed packages? Explain the concept of factors in R. How are they different from character vectors?		CO1
UNIT II			
Q3	Attempt any two parts	(5*2=10)	CO Mapping
a)	Compare and contrast matrices, arrays, and data frames in R. Provide examples of situations where each would be preferred. Write a function that converts a data frame into a matrix and vice versa, ensuring type safety.		CO2
b)	Explain the differences between cbind, rbind, and merge in R. Provide examples where each would be used. Write a function that combines multiple data frames by rows and columns, ensuring column name consistency.		CO2
c)	Write a custom function in R that automates the process of loading multiple CSV files from a directory, combines them into a single data frame, and performs basic data cleaning operations (e.g., handling missing values, renaming columns).		CO2